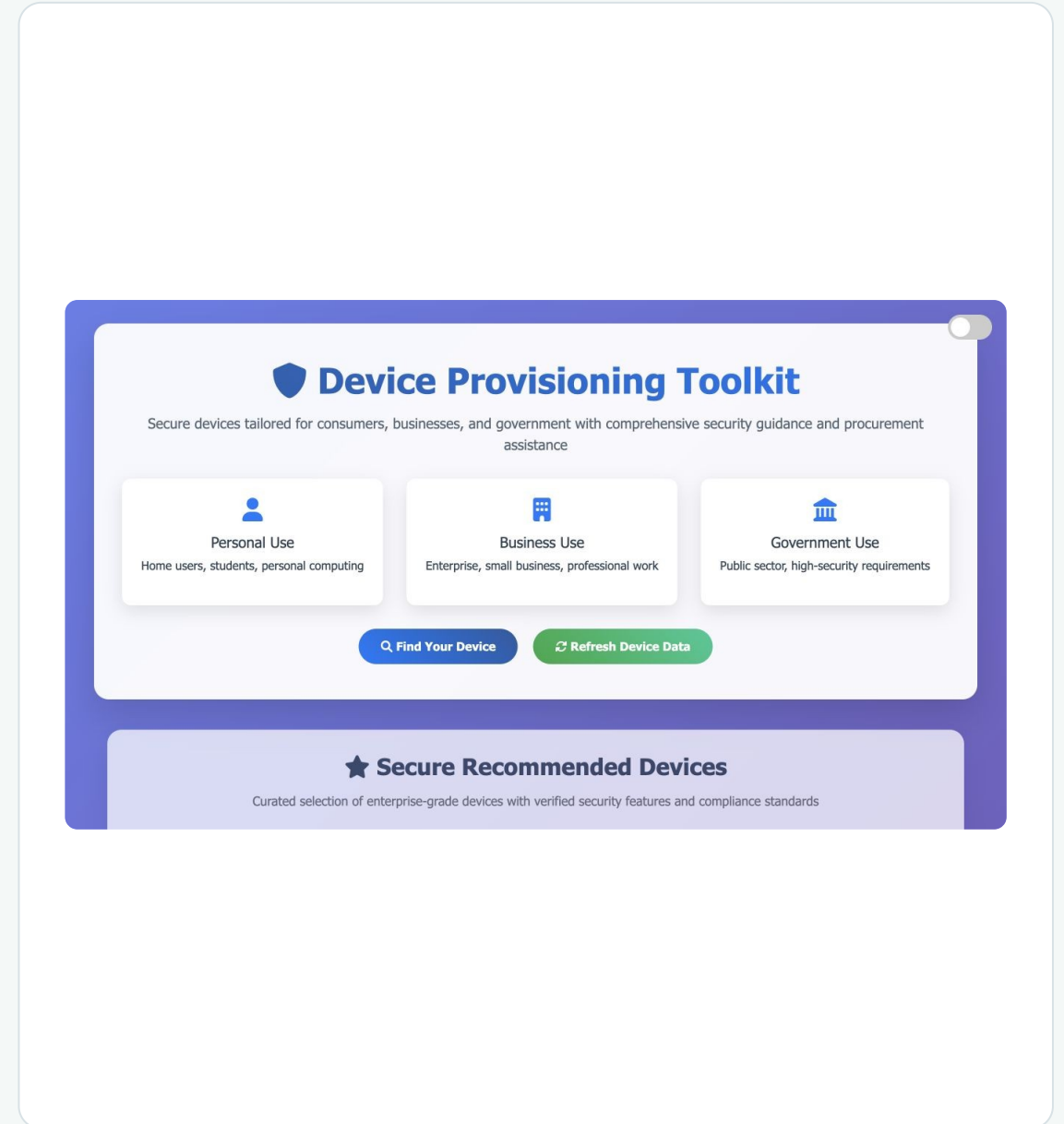


How do we decide what devices we provide?

A security-focused comparison and provisioning toolkit built from the dissertation.



The buying decision was missing a security standard



**FACULTY OF SCIENCE, ENGINEERING
AND COMPUTING**

School of Computer Science and Mathematics

**BSc (Hons) DEGREE
IN**

Cyber Security and Digital Forensics
Nathan Brown-Bennett

"How do we decide what devices we provide?"
18th July 2024

David Pickard

Kingston University London

The dissertation examined how governments, schools, employers and consumers choose devices—and whether security, practicality and reliability are considered together.

The gap

Guidance existed in fragments. Buyers could compare brand, price and specifications, but had no single workflow that translated security research into a usable decision.

Source: BSc dissertation, 18 July 2024

Three gaps kept recurring across the research



Security arrives late

Device decisions often begin with price, brand and familiarity; security is added after selection.



Guidance is fragmented

Hardware risks, software hardening and user capability are rarely explained in one place.

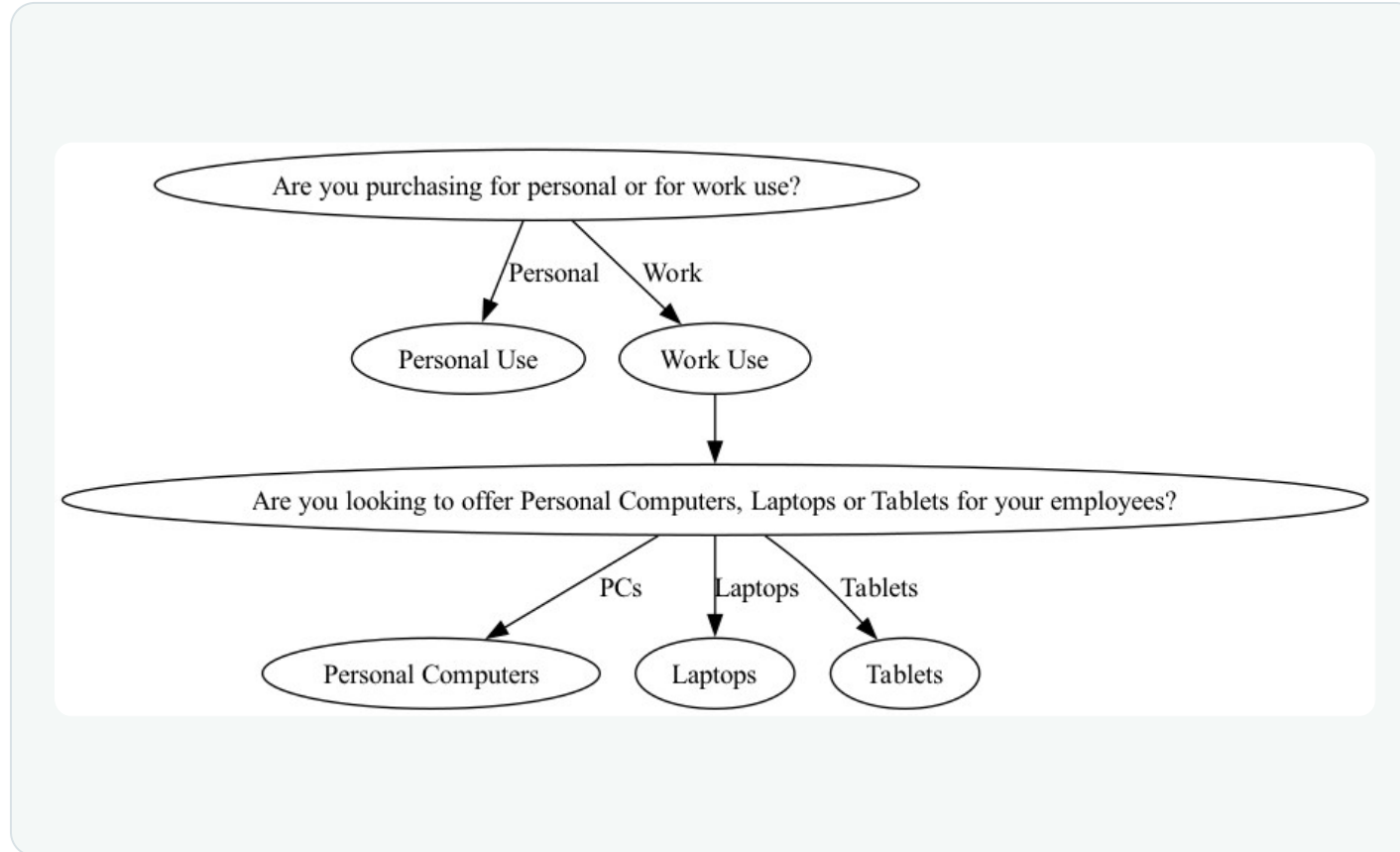


People need usable advice

A technically correct recommendation still fails if buyers cannot understand or apply it.

The product opportunity was not another comparison list. It was a decision tool that could connect research to action.

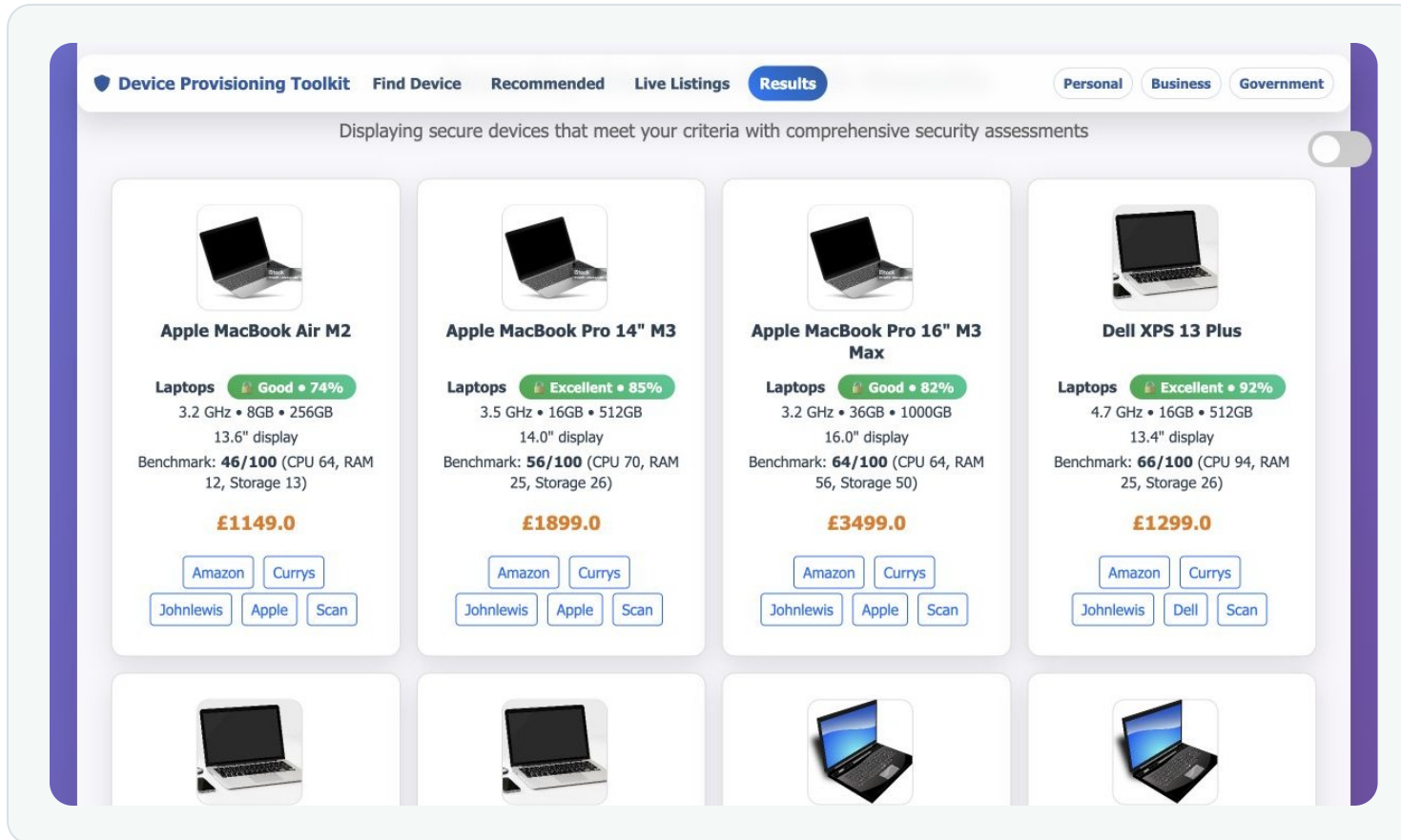
Research became a repeatable provisioning framework



The framework connects

- Buyer and user needs
- Hardware security considerations
- Operating-system and software controls
- Practicality and maintainability
- Clear recommendations

The current toolkit makes the research usable



What exists now

A working Flask application with device discovery, structured comparisons, security guidance, purchase context and provisioning recommendations.

Python • Flask • SQLite • Graphviz • Pandas

Search starts with the person, not the specification sheet

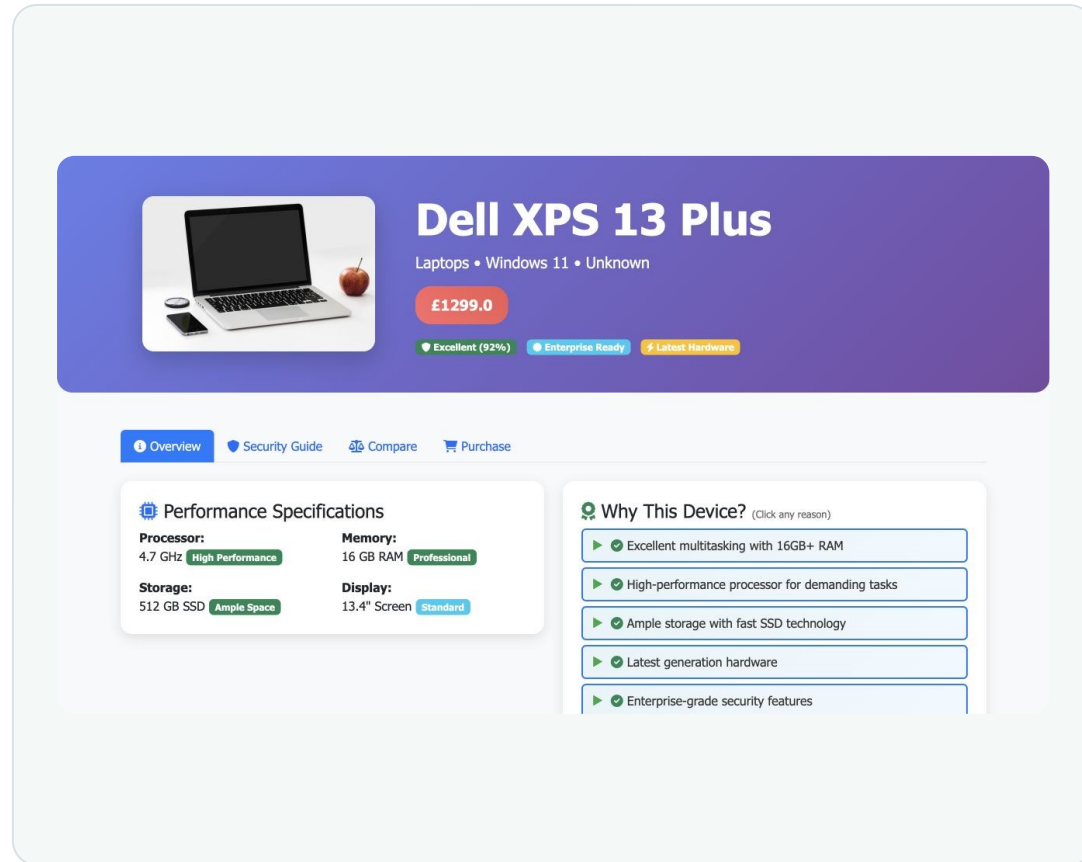
The screenshot shows a web interface for finding secure devices. At the top, there's a navigation bar with 'Find Device', 'Recommended', 'Live Listings', and 'Results' buttons. Below this is a search bar with the placeholder text 'Find device' and a hint 'Search by name, brand, or model...'. There are also filters for 'Device Type' (set to 'All Types') and 'Operating System' (set to 'Any OS'). A 'Preferred Brand' filter is set to 'Any Brand'. A price filter is set to '£1500'. A 'Show Advanced Filters' button is visible. At the bottom, there's a large blue button labeled 'Find Secure Devices' and a 'Close' button.

Describe the real use case

The screenshot shows a web interface for finding secure devices. At the top, there's a navigation bar with 'Find Device', 'Recommended', 'Live Listings', and 'Results' buttons. Below this is a search bar with the placeholder text 'Find device' and a hint 'Search by name, brand, or model...'. There are also filters for 'Device Type' (set to 'All Types') and 'Operating System' (set to 'Any OS'). A 'Preferred Brand' filter is set to 'Any Brand'. A price filter is set to '£1500'. A 'Show Advanced Filters' button is visible. At the bottom, there's a large blue button labeled 'Find Secure Devices' and a 'Close' button.

Refine the technical requirements

A device page explains the decision and the alternatives



The screenshot shows the product page for the Dell XPS 13 Plus. The header features a laptop image, the product name "Dell XPS 13 Plus", and the price "£1299.0". Below the price are three badges: "Excellent (92%)", "Enterprise Ready", and "Latest Hardware". The page includes a navigation bar with "Overview", "Security Guide", "Compare", and "Purchase". The main content is divided into two sections: "Performance Specifications" and "Why This Device?".

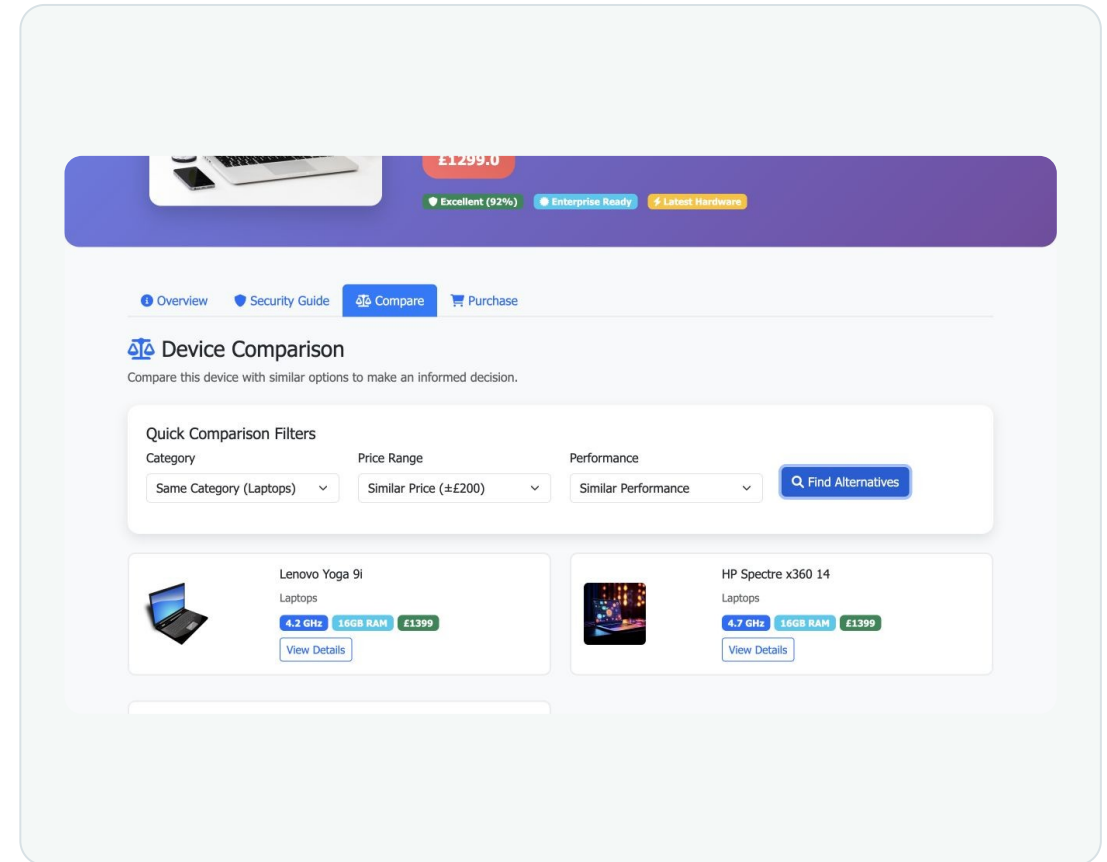
Performance Specifications

Processor:	Memory:
4.7 GHz High Performance	16 GB RAM Professional
Storage:	Display:
512 GB SSD Ample Space	13.4" Screen Standard

Why This Device? (Click any reason)

- ▶ Excellent multitasking with 16GB+ RAM
- ▶ High-performance processor for demanding tasks
- ▶ Ample storage with fast SSD technology
- ▶ Latest generation hardware
- ▶ Enterprise-grade security features

Specifications and reasoning



The screenshot shows the "Device Comparison" section of the product page. It includes a navigation bar with "Overview", "Security Guide", "Compare", and "Purchase". The main heading is "Device Comparison" with a subtext "Compare this device with similar options to make an informed decision." Below this is a "Quick Comparison Filters" section with three dropdown menus: "Category" (Same Category (Laptops)), "Price Range" (Similar Price (±£200)), and "Performance" (Similar Performance). A "Find Alternatives" button is next to the filters. The results section displays two alternative laptops: "Lenovo Yoga 9i" and "HP Spectre x360 14". Each alternative shows its specifications (4.2 GHz, 16GB RAM, £1399) and a "View Details" button.

Quick Comparison Filters

Category: Same Category (Laptops) | Price Range: Similar Price (±£200) | Performance: Similar Performance | **Find Alternatives**

Lenovo Yoga 9i
Laptops
4.2 GHz | 16GB RAM | £1399
[View Details](#)

HP Spectre x360 14
Laptops
4.7 GHz | 16GB RAM | £1399
[View Details](#)

Comparable alternatives

Selection leads to setup and procurement guidance



The screenshot displays the Dell XPS 13 Plus product page. The top section features a laptop image, the product name 'Dell XPS 13 Plus', and specifications 'Laptops • Windows 11 • Unknown'. The price is listed as £1299.0. Below this, there are three badges: 'Excellent (92%)', 'Enterprise Ready', and 'Latest Hardware'. The navigation bar includes 'Security Guide', 'Compare', and 'Purchase'. The 'Security Guide' section is expanded, showing 'Complete Security Setup Guide' with steps to ensure the device is properly secured before deployment. It includes 'Initial Security Settings' (Full Disk Encryption, BitLocker) and 'Recommended Security Software' (Enterprise Antivirus, Password Management).

Dell XPS 13 Plus
Laptops • Windows 11 • Unknown
£1299.0
Excellent (92%) Enterprise Ready Latest Hardware

Security Guide Compare Purchase

Complete Security Setup Guide

steps to ensure your device is properly secured before deployment.

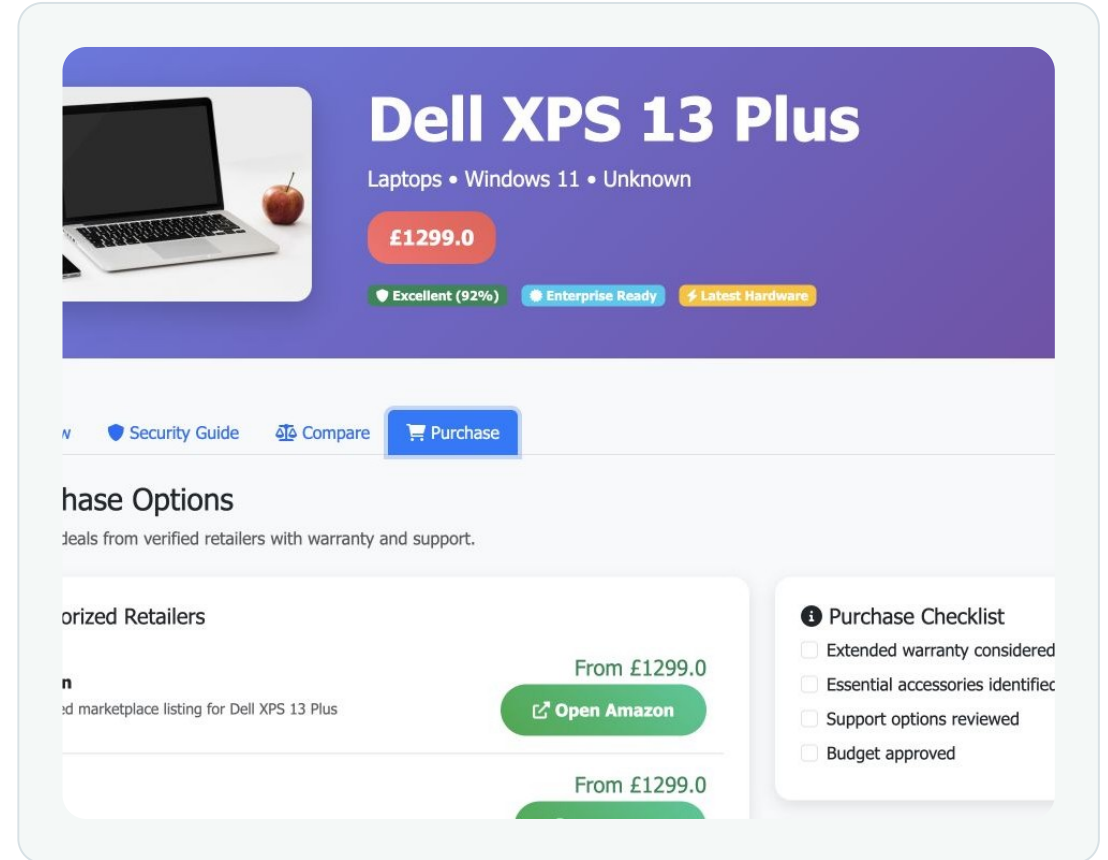
Initial Security Settings

- Full Disk Encryption
Locker in Control Panel > System and Security
- Require Strong Authentication
Windows Hello/Touch ID and require complex passwords

Recommended Security Software

- Enterprise Antivirus
 - Windows Defender (Built-in) - Recommended
 - Bitdefender GravityZone - Enterprise
 - CrowdStrike Falcon - Advanced
- Password Management

Secure the device before deployment



The screenshot displays the Dell XPS 13 Plus product page. The top section features a laptop image, the product name 'Dell XPS 13 Plus', and specifications 'Laptops • Windows 11 • Unknown'. The price is listed as £1299.0. Below this, there are three badges: 'Excellent (92%)', 'Enterprise Ready', and 'Latest Hardware'. The navigation bar includes 'Security Guide', 'Compare', and 'Purchase'. The 'Purchase Options' section is expanded, showing 'Authorized Retailers' with a link to 'Open Amazon' and a price of 'From £1299.0'. A 'Purchase Checklist' is also visible on the right side of the page.

Dell XPS 13 Plus
Laptops • Windows 11 • Unknown
£1299.0
Excellent (92%) Enterprise Ready Latest Hardware

Security Guide Compare Purchase

Purchase Options

Deals from verified retailers with warranty and support.

Authorized Retailers

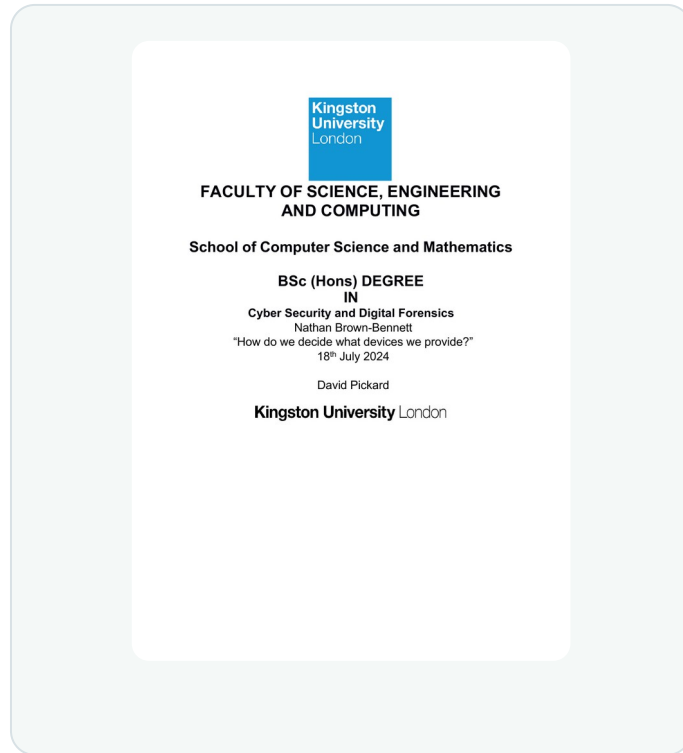
- Amazon
Amazon marketplace listing for Dell XPS 13 Plus
From £1299.0
Open Amazon
- From £1299.0

Purchase Checklist

- ☐ Extended warranty considered
- ☐ Essential accessories identified
- ☐ Support options reviewed
- ☐ Budget approved

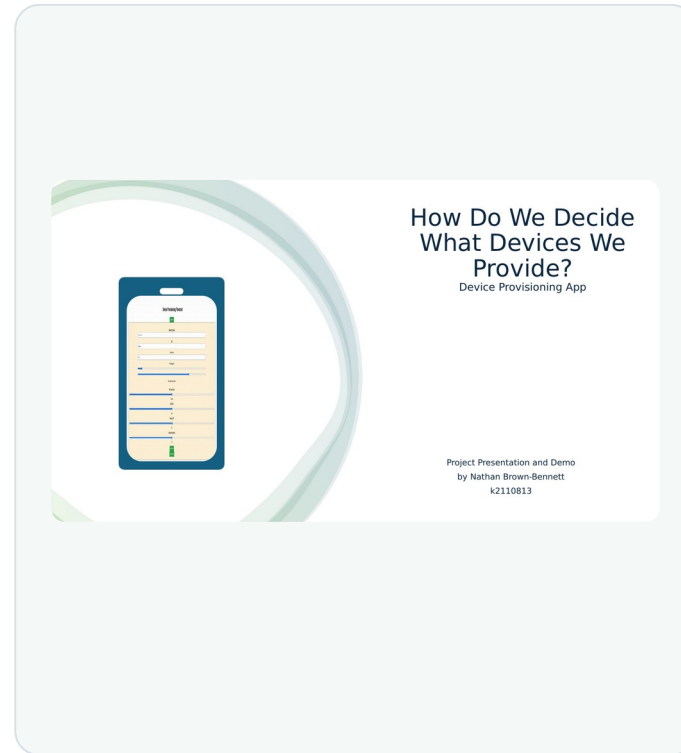
Complete the buying decision

The strongest evidence is the progression from question to product



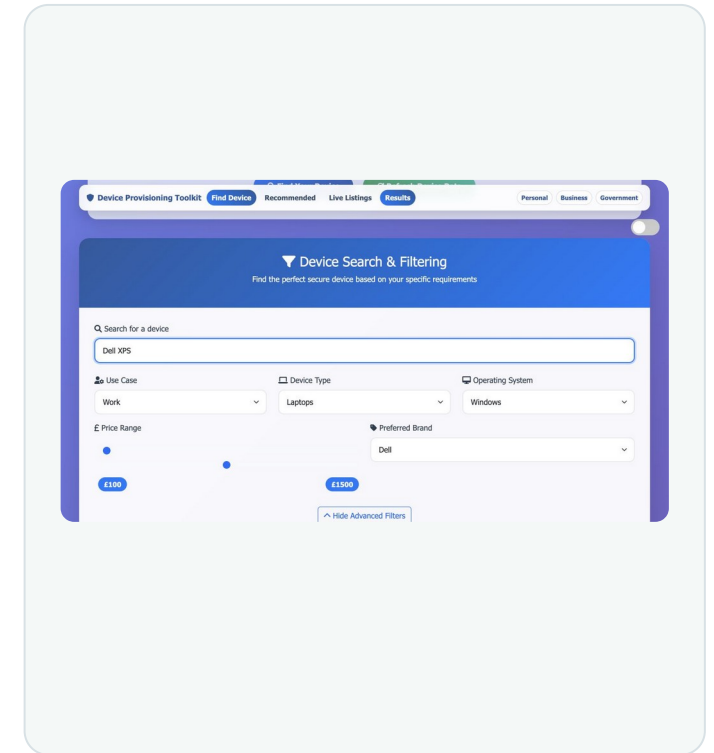
2024

Dissertation



2024

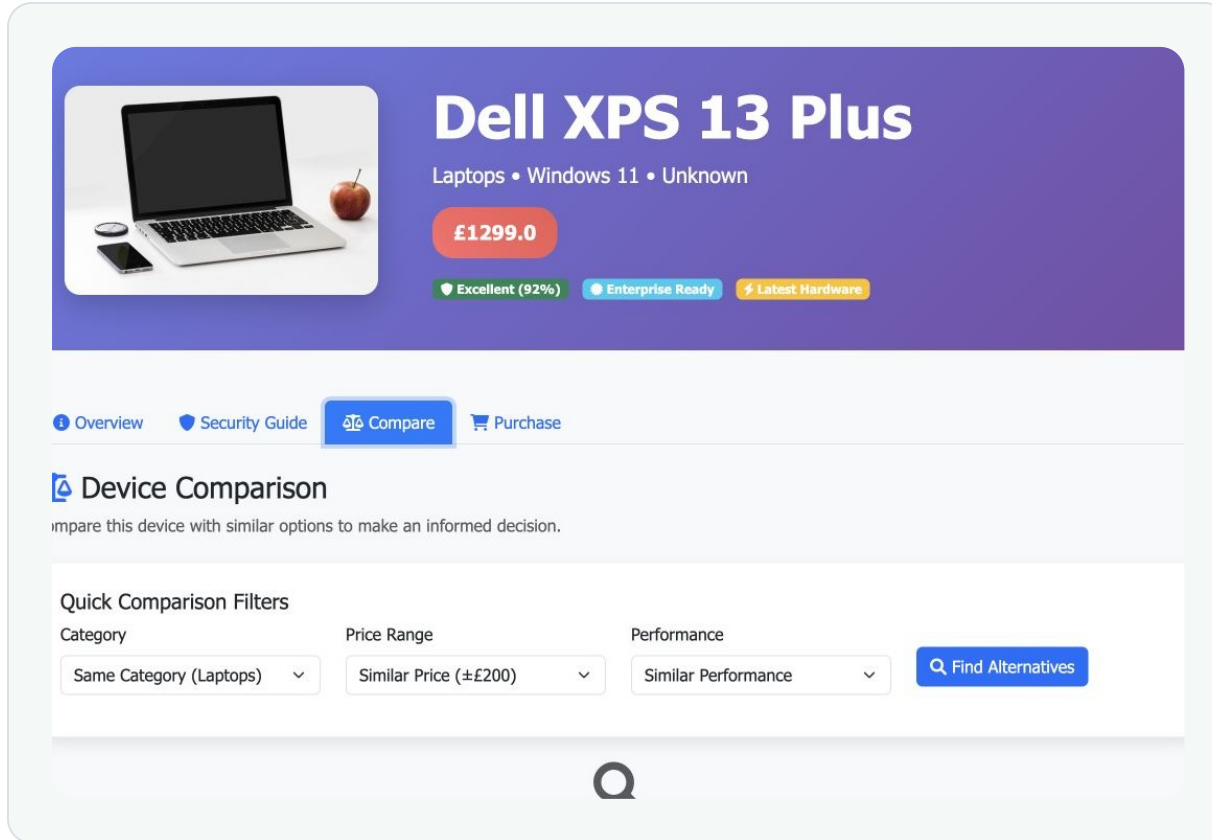
Original presentation



Current

Working toolkit

A lightweight stack keeps the working prototype portable



Product architecture

Python + Flask
Application and routing

SQLite + Pandas
Structured device evidence

Graphviz
Decision and provisioning flows

The next phase is stronger evidence, scoring and organisational validation




Broader evidence

Expand and maintain the device dataset with clearer provenance.



Explainable scoring

Make every recommendation traceable to transparent criteria.



Operational validation

Test workflows with procurement, IT support and security teams.

The product is already useful as a demonstrator; the opportunity is to make its evidence and operation durable enough for real organisations.

I build what should exist

This project started with a missing standard and became a working tool that makes device security decisions easier to understand and act on.

Dissertation

nathanbrownbennett.github.io/NathanBrown-Bennett/assets/documents/device-provisioning-dissertation.pdf

Repository

github.com/NathanBrownBennett/Dynamic-Flowchart-For-Device-Provisioning

